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Explainable Ensemble-Based Detection of Cyber Attacks on Internet of Medical Things

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***Abstract*—As new applications of the Internet of Things emerge in the health and medical services, malicious actors target these applications increasingly. These growing application bring a variety of privacy and security challenges. In this paper, we present an explainable machine learning ensemble designed to detect attacks on Internet-of-Medical-Things with high accuracy. The proposed system was tested using WUSTL-EHMS-2020 dataset. Tests showed that the proposed ensemble is capable of delivering an accuracy exceeding 99%, with an F_1 score exceeding 0.99. The proposed system was explained using SHAP values to provide insights into the most impactful features, and the nature of their impact on the system's decisions.**

***Index Terms*—iomt,iot,ml,machine learning,intrusion detection,explainable ml,xai**

I. Introduction

The adoption of the Internet-of-Medical-Things (IoMT) has seen rapid growth and this trend is only predicted to continue. According to [1], by 2022, electronic health devices are expected to hit a revenue of US$12.97bn, while the revenue of the eHealth Device market is expected to expand approximately 50% in the coming five years, as shown in figure 1. However, this increase in market expansion also means an increase in the number of cyberattacks from malicious actors.

IoMT devices contain sensitive private information and attack to these devices can be life-threatening. In comparison, a false data injection attack against a IoMT device has a completely different impact than a similar attack on smart home Internet-of-Things (IoT) device. When a false-data injection attack occurs in a smart home IoT context, such as a smart-home based thermometer, the air-conditioning system may operate in an unwanted way, causing discomfort to the owner. However, when the same attack occurs against an IoMT device like a thermometer attached to a patient, the smart system could be manipulated into dispatching the incorrect dosage of medication, thus, resulting in serious health consequences for the patient including death. The risk levels in IoMT context is many folds higher than other IoT contexts.

Although a number of different attacks on IoMT devices can cause harmful consequences, Man-In-The-Middle (MITM) attacks are of particularly dangerous. In a MITM attack, the attacker operates in the middle of two parties to intercept communication. Here, the attacker can impersonate a server, intercept traffic and send falsified commands to the IoMT device [2] otherwise known as a spoofing attack. Alternatively, they could also leverage their position to intercept commands issued from the sever, preventing the actual IoMT device from performing its intended purpose. In either scenario, a successful MITM attack could become life threatening for the patient.

In this paper, we present an explainable Intrusion Detection System (IDS) that uses network-flow features to identify attacks on IoMT environments. Based on voting ensemble-learning, the system can process network-extracted information to detect, and block attacks, and protect IoT devices from malicious traffic. The proposed system is explained using Shapley Additive Explanation (SHAP) values to provide insights into the impact of each feature on the classifier's decision. This establishes better trust in the system, and ensures that the decisions are not originating from a black box.

In the next section of this paper, we will discuss the relevant literature on the topic. Section III introduces the proposed system while Section IV covers the details around the selected dataset, its preprocessing and the implementation environment. Experiment descriptions and their results are detailed in Section V. The proposed model's explainability is discussed in details in Section VI. Finally, a discussion of the results and a comparative analysis can be found in Section VII and the conclusions and future work are documented in the final section of this paper.

II. Related Works

Recently, IoT Intrusion Detection research has seen a spike in popularity thanks to a number of realistic and synthetic datasets that captured traffic like Bot-IoT [3], TON_IoT [4], and IoT-ID [5]. However, the available IoT IDS research demonstrates a clear absence of realistic datasets that cover attacks and normal traffic data, especially in the medical field. Thus, previous research fails to consider the unique variables of medical IoT devices, including size, and number of packets, data seize generated and data sensitivity. Our selected dataset,

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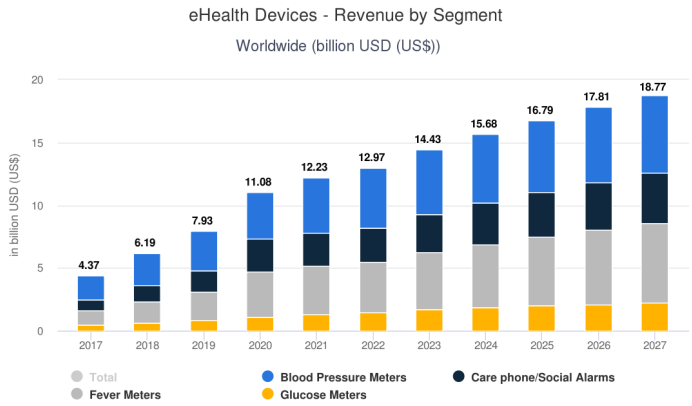

Fig. 1: Historical eHealth Device Market Revenue by Segment with Future Projections [1]

WUSTL-EHMS-2020, effectively captures medical IoT data. In this section, we will review key IoMT research.

Hady et al. proposed the WUSTl-EHMS-2020 dataset [6] using a k-Nearest Neighbor (kNN) model and a machine learning-based intrusion detection system that assessed multiple classifier algorithms. Their architecture demonstrated a successful accuracy of 91.56%, and AUC of 0.8748. Low AUC impacted the classifiers, likely because of the class imbalance issue that was neglected in the dataset.

In 2021, Lee et al. presented the M-IDM [7], a multi-class classification-based IoT IDS for the medical environment. Using Convolutional Neural Networks (CNN), the model split the data into class types, including critical, informal, major, and minor, for intrusion detection. The architecture produced an AUC of 0.967 and an F_1 score of 0.937, with and accuracy of approximately 94%.

In 2022, Gupta et al. presented an IoMT network IDS system based on a tree classifier [8]. With the help of GridSearchCV and the WUSTL-EHMS-2020 dataset, they were able to develop a tree classifier-based intrusion detector that promotes dimensionality reduction and increases the detection rate. The proposed system was tested, and performed with an accuracy of 94.23% with an F_1 score of 0.938.

In 2022, Wahab et al. proposed a machine learning based intrusion detection system for IoMT devices that used a hybrid model consisting of long short-term memory (LSTM) and gated recurrent unit (GRU). Their work utilized a CICDDoS2019 dataset [9] to test and train the research. While this dataset captured Distributed Denial of Service (DDoS) attacks on computers and performed 99.01%, the dataset itself is not an IoT or IoMT dataset, thus, the research fails to accurately address IoMT security issues.

In 2022, Nandy et al. proposed an IoMT intrusion detection system based neural networks optimized by swarm intelligence [10]. The proposed dataset was TON_IoT. Like Wahab et al., the architecture trained and tested the research using a non-IoMT dataset and therefore, failed to gather any research on IoMT traffic. The produced high accuracy of 99.5% is not representative of the quality of detection for IoMT traffic.

In 2023, Alani et al. proposed an explainable deep neural network-based intrusion detection system for IoMT networks named XMeDNN [11]. The proposed explainable system utilized a multi-layer perceptron architecture, and was trained using WUSTL-EHMS-2020 dataset. XMeDNN delivered an accuracy of 96.87%, with an F_1 score of 0.968.

Further information on research in IoMT security and machine-learning can be found in [12].

III. Proposed System

In this paper, we proposed a ensemble-based intrusion detection system that is trained to detect attack in IoMT environments. Figure 2 shows an overview of the proposed system.

As shown in the figure, the training stage includes preprocessing, training of a pipeline of different classifiers to create the ensemble. The proposed ensemble model utilizes a random forest (RF) classifier, a decision tree (DT) classifier, a support vector machine (SVC) classifier, and an extreme gradient boost (XGB) classifier. These four classifier are used to build a voting ensemble to enhance the accuracy of the proposed system [13]. The voting method chosen was "soft" voting. In *soft voting* the prediction of the ensemble is based on summing the probabilistic prediction produced by the individual classifiers within the ensemble. Once the training stage is completed, the trained ensemble model is then sent to the deployment stage.

At the deployment stage, the network traffic is captured as raw packets. These raw packets are then passed through a network-flow analysis package named ARGUS [14]. This tool, extracts network-flow features from raw captured traffic. These features are then passed as the input to the trained DNN model for inference. The trained ensemble-model then produces a prediction whether the captured network flow is normal or attack traffic. This prediction can later be linked to a firewall to block the source IP address and stop the traffic.

IV. The Dataset

The dataset chosen for the training and testing of our proposed system is WUSTL-EHMS-2020 [6]. IoMT devices

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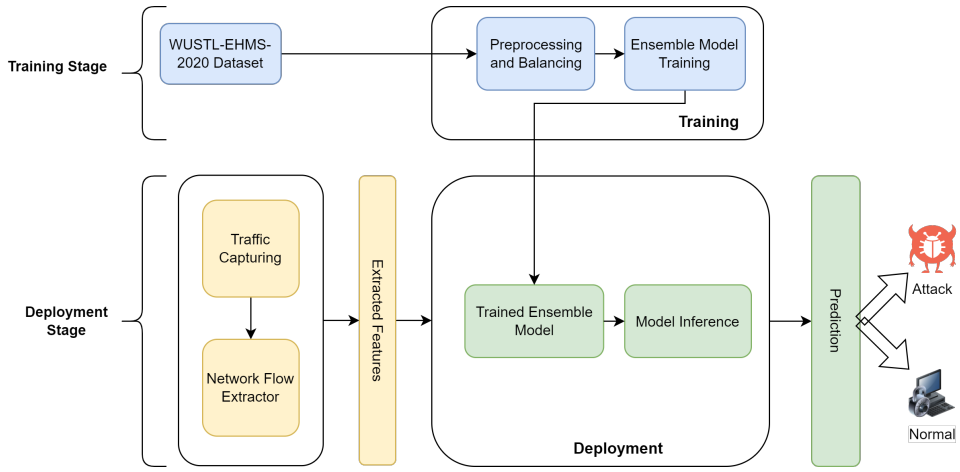

Fig. 2: An overview of the proposed system

SpO_2 measurement device, blood pressure measurement device, Electro-Cardio-Graph (ECG) device, and a thermometer were connected in a testbed and employed to build the dataset. Network traffic and device readings were both collected to create the dataset. With the help of ARGUS, a network flow extraction tool, the traffic files were processed effectively to generate the dataset of network-flow information.

The original dataset was composed of 16,138 samples (14,272 normal, and 2,046 attack), each with 35 network-flow features, and eight device reading features. The device biometric readings were consequently removed to align with the purpose of our model to detect attack through the analysis of network traffic.

A. Implementation Environment

The data preprocessing, model training, and testing were performed in an environment with the following specifications:

- Processor: AMD Ryzen 5 3600 (4.2GHz)
- RAM: 128GB
- Operating System: Windows 10 Pro
- Python v3.10.0
- Sci-Kit Learn v1.1.3
- Matplotlib v3.5.3
- SHAP v0.42.0

B. Preprocessing

Upon close examination of the dataset, we made the following observations:

1) A noticeable imbalance was detected between the "normal" class and the "attack" class, with the attack class making up only 12.5% of the total number of samples.
2) The dataset features include many host-specific features such as source and destination IP addresses, and source and destination MAC addresses.
3) The dataset includes two samples with invalid data (in the form of text in features that are numerical).

Based on these observations, the preprocessing steps shown in Algorithm 1 was performed.

Algorithm 1: Dataset Preprocessing

Input: Original dataset
Output: Preprocessed dataset

$Array \leftarrow Dataset$
$Array \leftarrow Array(remove_unwanted_features)$
for *i in 16,318* **do**
  if *Array(i) = invalid_data* **then**
    Remove_Sample Array(i)
  end
end
$Array \leftarrow RandomOverSampling(Array)$
$Dataset \leftarrow Array$

To effectively begin the preprocessing of the dataset, unwanted features, including those that were host-specific such as IP and MAC addresses were removed. This step resulted in the removal of seven features, with 28 features remaining. This removal supports classifier generalization and performance when tested with samples from outside of its training subset.

Next, the two samples with invalid data were removed, leaving 16,316 samples (14,270 normal, and 2,046 attack).

Finally, random over-sampling of the minority class (attack) was performed to address the imbalance between the normal and attack samples. After this step, the two classes had an equal number of samples, resulting in a dataset with 28,540 samples (14,270 normal, and 14,270 attack), with 28 features.

V. Experiments and Results

The data was randomly split into 75% training subset, and 25% testing subset. The random split took into account maintaining the balance between the normal and attack classes through stratified split.

A. Experiment Setup

At the first step of experiments, we build the four classifiers that will form the basis of the proposed ensemble classifier.

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TABLE I: Classifiers Hyperparameters

Classifier	Hyperparameter	Value
RF	Trees	100
	Criterion	Gini Impurity
	Min Sample Split	2
	Max Features	Sqrt
DT	Criterion	Gini Impurity
	Splitter	Best
	Min Sample Split	2
	Min Impurity Decrease	0.0
SVC	Kernel	RBF
	Degree	3
	Gamma	auto
	Tolerance	10^{-3}
XGB	Learning Rate	0.3
	Min Split Loss	0
	Max Depth	6
	Max Delta Step	0

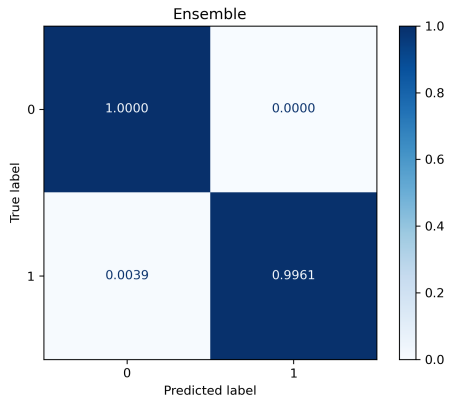

Fig. 3: Confusion matrix plot for the ensemble classifier

The four classifiers; RF, DT, SVC, and XGB were trained using the training subset that represents 75% of the samples randomly selected with stratified split. The trained four classifiers were then tested using the remaining 25% of samples. The hyperparameters of the four classifiers are shown in Table I.

In the soft voting ensemble used in the proposed system, each classifier will produce a probabilistic prediction that indicates the probability of the inputs sample being classified into a specific class. Then, the ensemble sums these probabilities to decide which class would the final prediction be.

Each of these classifiers were trained using the training subset, and tested with the testing subset. The results of testing can be found in Table II.

As we examine the results in the table, we can see that the proposed ensemble architecture slightly outperforms SVC classifier with F_1 score of 0.9980, and they both noticeably outperform the other classifiers. Figure 3 shows the confusion matrix plot of the proposed ensemble classifier. The confusion matrix plot shows that the proposed classifier achieves a false-positive rate of 0%, and a false-negative rate of 0.39%.

Another important performance metric is the area under the curve (AUC) in the receiver operating characteristic (ROC)

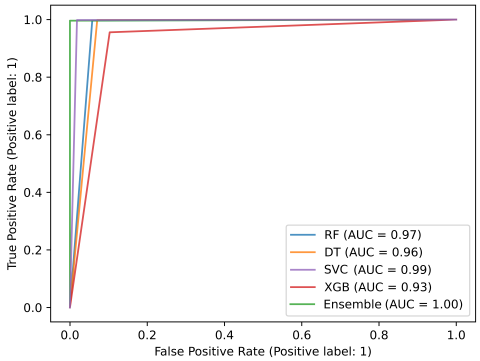

Fig. 4: AUC ROC curve for the five classifiers

curve. This curve measures the area under the curve for the binary classifier where the axis of the curve are the FPR and TPR. Figure 4 shows the AUC ROC curve for all of the five classifiers. This figure shows that ensemble classifier outperforms the original four classifiers used in constructing it, with area under curve of 1.00.

B. 10-Fold Cross-Validation

To ensure that the proposed classifier can generalize well, we performed 10-fold cross-validation. In a 10-fold cross-validation, the dataset is randomly divided into ten different subsets. Ten training cycles are performed where one of the subsets is used for testing and the remaining nine are used for training. At the end of the 10 cycles, each one of the 10 subsets was used for testing, while excluded from training, once. At the end of the cross-validation process, if the results of the 10 folds have low standard deviation, the mean value of the performance metrics can be considered a reliable result. The results of applying 10-fold cross validation on our proposed model can be found in Table III.

By examining the results shown in the table, it is clear that the standard deviation is minimal, and the results of the 10 folds are very aligned with each other, with an average accuracy of 0.999615, and an average F_1 score of 0.999618. This result establishes that the proposed classifier generalizes well beyond its training dataset.

VI. Model's Explainability

Research on explainable machine learning has gainied a lot of momentum in the past few years, with cybersecurity being one of the fields that could significanlty benefit from explainability [15]. As the number of attacks and malicious actors rapidly increases, defenders need all the possible support. Machine learning presents itself as a possible path to address the ever increasing attack surface [16]. Applications of machine learning in cybersecurity expanded in many directions, such as intrusion detection [17], malware detection [18], phishing detection [19], botnet detection [20], among many other applications. However, these applications has always faced reluctance by cybersecurity practitioners. Explainability helps in providing a deeper understanding how the decisions

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TABLE II: Testing results for all classifiers

	Accuracy	Precision	Recall	F_1 Score	Training Time (S)	Testing Time (μs)
RF	0.969306	0.970687	0.969302	0.969284	12.48005	53.900175
DT	0.963980	0.966219	0.963976	0.963937	0.075566	0.087481
SVC	0.990049	0.990184	0.990048	0.990048	250.207342	630.851049
XGB	0.926419	0.927891	0.926415	0.926355	0.380673	0.194010
Ensemble	0.998038	0.998045	0.998038	0.998038	267.20128	686.551043

TABLE III: Results of the 10-fold cross-validation for ensemble classifier

Fold	Accuracy	Precision	Recall	F_1 Score
Mean	0.999615	0.999932	0.999303	0.999618
St Dev	0.000428	0.000203	0.000824	0.000424

TABLE IV: SHAP feature importance

Feature	Importance	Feature	Importance
DIntPkt	0.083558	TotBytes	0.000185
DstJitter	0.080674	pLoss	0.000031
SrcLoad	0.074083	pSrcLoss	0.000022
Load	0.067491	Loss	0.000013
DstLoad	0.065956	sMaxPktSz	0.000009
Sport	0.056579	pDstLoss	0.000008
Rate	0.043453	SIntPktAct	0.000005
SrcJitter	0.035816	dMaxPktSz	0.000001
SIntPkt	0.020046	DstGap	0
TotPkts	0.000386	Trans	0
DstBytes	0.000339	Dport	0
Dur	0.000326	DIntPktAct	0
SrcBytes	0.000299	SrcGap	0
sMinPktSz	0.000195	dMinPktSz	0

are made in ML-based classifiers and how each feature impacts the decision. This kind of transparency increasses trust in these systems, and ensure that their decisions do not originate from a black box state.

SHAP was first presented in [21] as a model-agnostic explanation method. SHAP explanations are based on Shapley values, originally presented in game theory. The contribution of each feature to the decision made by a ML-based model is calculated by measuring the model's performance with and without this feature. Table IV shows the calculated absolute average feature importance for all features within the dataset.

As shown in the table, the features `DstGap, Trans, Dport, DIntPktAct, SrcGap, dMinPktSz` have shown an importance of zero. This means that the values of these features have no impact on the prediction decision.

Figure 5 shows the SHAP values summary plot for the top five features impacting the classifier's decisions. Each point in the plot represent one feature value in one data sample. Points on the right side of the plot draw the prediction closer to a higher value which makes it closer to a 'malicious' prediction, while points on the left side draw the prediction to a lower value brining the prediction closer to 'benign'. Red dots represent higher value of the feature, while blue dots represent lower value of the features.

As shown in the figure, the feature with the highest impact is `DIntPkt`, which represents the destination packet inter-arrival time measured in milliseconds. The figure shows that higher inter-arrival times push the prediction closer to malicious, while lower values push the prediction closer to benign. A similar behavior is shown in the second feature in the figure; `DstJitter`, which represents the jitter in the arrival time of the packets. These two features are logically correlated and hence, have similar impact on the prediction decision. This explanation is aligned with the fact that attack packets are often not sent in rapid-fire fashion, but as individual packets with high time spacing.

The features `SrcLoad, Load, DstLoad` ranked third, fourth, and fifth, consecutively, in terms of impact on the prediction. These three features are also correlated as the `Load` is calculated from `sLoad`, and `dLoad`. In general, the load in the dataset is calculated as the data rate in bits per second. Figure 5 shows that high load does not always indicate malicious traffic. However, the figure shows a range of low values being impactful in deciding that the traffic is malicious. This is also aligned with the fact that most attacks that are not targeting availability, have data rates that are lower than normal traffic. This applies to many attack types including injection attacks, and scanning attacks.

VII. Discussion

The results obtained in the testing stage were validated by performing a successful 10-fold cross-validation process. This validation proves that the proposed system can generalize well beyond its training dataset. In addition, these results have shown that the trained ensemble classifier outperforms outperforms ML-classifiers in terms of performance metrics.

Table V shows a comparison of the results obtained with several previous works. Within this comparison, research papers that did not utilize IoMT datasets were excluded, as their results are considered not representative of IoMT-based systems.

As shown in the table, our proposed system outperforms all related works by a good margin in terms of accuracy, F_1 score, as well as the false-positive rate. The comparison, in addition to the 10-fold cross-validation, shows that the results of our proposed system are reliable.

VIII. Conclusions and Future Work

In this paper, we presented an ensemble-based intrusion detection system for IoMT environments. The proposed system was trained and tested using the WUSTL-EHMS-2020 dataset. The proposed system delivered a testing accuracy of

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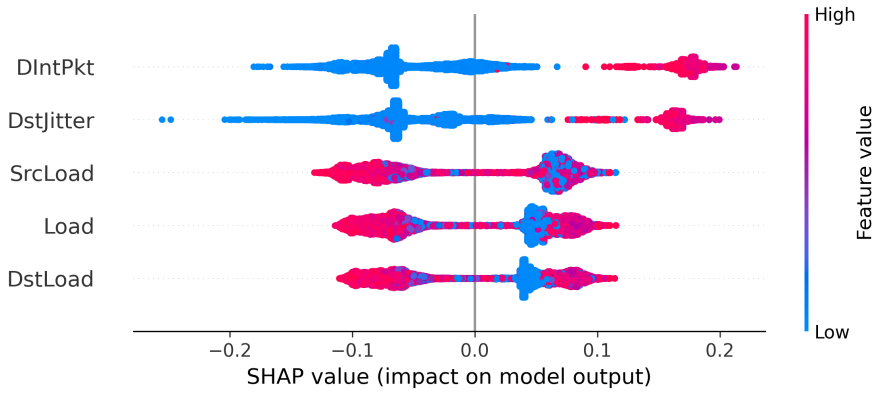

Fig. 5: Summary plot of the proposed model's SHAP values

TABLE V: Comparison of the results of the proposed system with related works

Research	Dataset	Classifier	Accuracy	F_1 Score	FPR
Gupta et al. [8]	WUSTL-EHMS-2020	Tree	94.23%	0.938	6%
Hady et al. [6]	WUSTL-EHMS-2020	kNN	91.56%	-	-
Lee et al. [7]	[7]	CNN	94%	0.937	5.3%
Alani et al. [11]	WUSTL-EHMS-2020	MLP	97.57%	0.976	3.1%
Proposed system	WUSTL-EHMS-2020	Ensemble	99.80%	0.9980	0%

99.80%, with an average accuracy of 99.96% in 10-fold cross-validation. Testing have also shown a F_1 score of 0.9980. The proposed system delivered better performance metrics when compared to related previous works, as discussed in Section VII.

Future directions of this research include the following.

1) Explore the possibility of specific attack identification, in addition to attack detection.
2) Explore the use of additional datasets for training and validation.
3) Explore the employment of feature selection mechanisms to improve the system's efficiency.

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